Method to the Madness

- Two datasets: **50m** and **80m** wind speeds
- Variables:
  - `y` = observed wind speed
  - `day` = numeric time index (not used in this model)
- Goal: model the *distribution of wind speeds* at each height using a Normal–Extreme Value mixture, estimate the mode parameter ξ, construct CIs, and test whether the latent wind distributions differ by height.

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1. What Is the Statistical Model in This Context?

You observe daily wind speeds:

- $Y_{50,i}$ = wind speed at 50m on day i
- $Y_{80,i}$ = wind speed at 80m on day i

We are **not modeling time trends** (`day` is unused here). Instead, we are modeling the **distribution of wind speeds across days**.

The Modeling Idea

Wind speed varies from day to day because:

1. There is a "true wind level" on each day (a latent mean), call it μ_i.
2. The observed wind speed fluctuates around that true level.

So we build a hierarchical model:

Step 1: Measurement model (within-day variation)

Given the day-specific latent level μ_i:

$$Y_i \mid \mu_i \sim N(\mu_i, 1).$$

Interpretation:

- Observed wind speed fluctuates normally around a latent daily wind level.
- Variance is fixed at 1 (given in problem).

Step 2: Distribution of daily wind levels

The latent daily wind levels follow a right-skew Extreme Value distribution:

$$\mu_i \sim \mathrm{EV}(\xi, \theta),$$

with density

$$g(\mu \mid \xi, \theta) = \frac{1}{\theta} \exp\left(-\frac{\mu - \xi}{\theta}\right) \exp\left[-\exp\left(-\frac{\mu - \xi}{\theta}\right)\right].$$

Interpretation:

- ξ = **mode** of the wind-level distribution.
- θ = scale (spread).
- Skewness reflects extreme-wind behavior (occasional unusually high wind regimes).

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2. What Distribution Does Wind Speed Follow?

Since μ_i is unobserved, the observed wind speed density is:

$$m(y \mid \xi, \theta) = \int_{-\infty}^{\infty} f(y \mid \mu) g(\mu \mid \xi, \theta) d\mu.$$

This is a **Normal location mixture with an Extreme Value mixing distribution**.

Why This Is Not a Simple Regression Problem

In an ordinary regression model, we would specify something like:

$$Y_i = \beta_0 + \varepsilon_i,$$

or possibly include covariates, and then directly estimate parameters from a closed-form likelihood.

Here, however:

- The mean μ_i is **unobserved**.
- The likelihood of Y_i is not directly Normal.
- We must integrate out the latent variable μ_i to obtain the marginal density.

Because of the integral

$$m(y \mid \xi, \theta) = \int f(y \mid \mu), g(\mu \mid \xi, \theta), d\mu,$$

there is:

- No closed-form expression for the likelihood,
- No linear estimating equations,
- No sufficient statistics like a sample mean or variance,
- No direct least-squares interpretation.

Each evaluation of the likelihood requires **numerical integration**.

This makes the model fundamentally a **hierarchical mixture model**, not a regression model. The parameters (ξ, θ) describe the *distribution of latent daily wind regimes*, not a linear mean relationship.

That is why your code must:

- Numerically integrate for every observation,
- Numerically maximize the likelihood,
- Use Hessians or profile likelihood methods for inference.

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3. What Are We Estimating?

For each height (50m and 80m separately), we estimate:

- $\xi \to$ the **most likely latent wind level**
- $\theta \to$ spread of latent wind levels

The key parameter of interest:

$$\xi = \text{mode of daily wind level distribution.}$$

This answers:

What is the typical (most probable) daily wind regime at this height?

Note: this is not the mean wind speed, but the most probable latent regime.

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4. Maximum Likelihood Estimation

For one dataset (say 50m), the log-likelihood is:

$$\ell(\xi, \theta) = \sum_{i=1}^{n} \log m(y_i \mid \xi, \theta).$$

Because $m(y_i \mid \xi, \theta)$ involves numerical integration, **each likelihood evaluation requires integration**.

You maximize this using `optim()` with parameterization:

$$(\xi, \log \theta)$$

to ensure $\theta > 0$.

This is fully numerical likelihood-based estimation.

Why the moment-based starting values?

For this model:

$$E(Y) = \xi + \gamma\theta$$

$$\mathrm{Var}(Y) = \frac{\pi^2}{6}\theta^2 + 1$$

So you approximately invert these to get good starting values.

This improves numerical stability during optimization.

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5. Confidence Intervals for ξ (Mode of Wind Regime)

You compute two types:

(A) Wald CI

From the Hessian matrix:

$$\hat{\xi} \pm 1.96, \mathrm{SE}(\hat{\xi}).$$

This assumes:

$$\hat{\xi} \approx N(\xi, \mathrm{Var}).$$

This works well if the likelihood is approximately quadratic near the maximum.

(B) Profile Likelihood CI

Instead of relying on a quadratic approximation, you:

1. Fix ξ
2. Re-optimize over θ
3. Solve

$$2(\ell_{\max} - \ell_{\mathrm{prof}}(\xi)) = \chi^2_{1,0.95}.$$

This approach is more robust when the likelihood is skewed.

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6. Comparing 50m and 80m

Now we test whether the latent wind distributions differ by height.

Full Model (different distributions)

50m has:

$$(\xi_{50}, \theta_{50})$$

80m has:

$$(\xi_{80}, \theta_{80})$$

Total parameters = 4.

Reduced Model (same distribution)

Both heights share:

$$(\xi, \theta).$$

Total parameters = 2.

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7. Likelihood Ratio Test

Test:

$$H_0 : \xi_{50} = \xi_{80}, ; \theta_{50} = \theta_{80}$$

vs

$$H_1 : \text{distributions differ.}$$

Test statistic:

$$\Lambda = 2(\ell_F - \ell_R).$$

Under H_0:

$$\Lambda \sim \chi^2_2.$$

If $p < 0.05 \rightarrow$ wind regimes differ by height.

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8. Graphing the Mixing Densities

You plot:

$$g(\mu \mid \hat{\xi}, \hat{\theta})$$

These curves represent:

> The distribution of daily latent wind levels at each height.

If curves are clearly separated:

- The typical wind regime differs by height.

If nearly identical:

- Same wind structure at both heights.

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9. Interpretation in Wind Context

This model says:

- Wind speeds fluctuate daily.
- Each day has a latent wind level.
- Those latent levels follow a skewed extreme-value distribution.
- The parameter ξ represents the **most likely daily wind level**.
- The parameter θ represents variability in wind regimes.
- The LRT determines whether 50m and 80m have fundamentally different wind regime structures.

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10. What Is the General Method Being Used?

This is a **likelihood-based hierarchical mixture modeling approach**:

Step 1

Specify conditional model:

$$Y_i \mid \mu_i$$

Step 2

Specify latent distribution:

$$\mu_i \sim g(\cdot)$$

Step 3

Integrate out latent variable:

$$m(y) = \int fg$$

Step 4

Maximize marginal likelihood numerically.

Step 5

Use likelihood theory for:

- Wald CI
- Profile CI
- Likelihood ratio test

Conceptually

You are asking:

> Are the *underlying wind regimes* at 50m and 80m statistically the same?

Not:

> Are individual wind speeds different?

This is a structural comparison of the entire latent wind distribution, and the required integral is precisely what makes it a hierarchical mixture problem rather than a simple regression.